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Practitioner Report

Behavioural activation for the treatment of low-income, African American adolescents with major depressive disorder: a case series

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Behavioural activation (BA) is a psychosocial treatment that has shown promise in the treatment of adults suffering from major depressive disorder (MDD). Recent studies have shown that BA may also be effective for treating depressed adolescents. There are no studies that have reported on the BA treatment of depressed and low-income African American adolescents; thus, the current study reports on the effectiveness of a version of BA adapted for the treatment of African American adolescents who were diagnosed with MDD ($n = 3$). Participants were allowed to attend a maximum of 17 sessions of weekly psychotherapy. Based on results taken from structured interviews, two of the three participants no longer met criteria for MDD at the end of treatment, and the severity of clinician-rated depressive symptoms and impairment decreased for all participants at post-treatment assessment. Additionally, all participants and their caregivers reported satisfaction with treatment. Implications of these findings, study limitations and suggestions for future directions are discussed.

Key Practitioner Message:

- Behavioural activation is an emerging evidence-based psychotherapeutic treatment for adolescent depression.
- African American adolescents in particular can benefit behavioural activation.
- Behavioural activation is an intervention that can be utilized in settings with limited resources.

Keywords: Major Depressive Disorder, Behavioural Activation, Adolescents, Treatment

Major depressive disorder (MDD) among children and adolescents occurs frequently, is often chronic and is associated with significant morbidity and mortality (Ryan, 2005). Of children and adolescents, 14–25% experience an episode of major depression before adulthood (Kessler & Walters, 1998; Lewinsohn, Rohde, & Seeley, 1998). In addition, the rate of depression in young people increases dramatically between ages 15 and 18 (Hankin et al., 1998). Depressive disorders in youth affect development, academic performance, and family and peer relationships, and are accompanied by a poorer long-term prognosis: depressed adolescents go on to have lower incomes as adults; are less likely to earn college degrees; and are more likely to become unwed parents, to have substance abuse problems and to report experiencing stressful events in their lives (Lewinsohn et al., 1998; Lewinsohn & Clark, 1999). Moreover, major depression is strongly correlated with completed suicide, which is the second leading cause of death for youth ages 15–19 (Kann et al., 2000; Brent, 2001; World Health Organization, 2010). For all of these reasons, it is important to develop early interventions for MDD in adolescents.

The current standard of care for treating MDD includes pharmacotherapy and cognitive–behavioural therapy (CBT; Birmaher, Brent, & American Academy of Child and Adolescent Psychiatry group, 2007). Because of the many side effects of medication management, including the black box warning about the potential increased risk of suicidal behaviour in teens taking certain antidepressants (United States Food and Drug Administration, 2007), practitioners often opt for psychotherapeutic interventions as first-line treatments for depression in adolescents (Michael & Crowley, 2002). Ample evidence supports the use of CBT for depression in both adult and adolescent samples (Klein, Jacobs, & Reinecke, 2007); however, recent evidence from the adult literature suggests that the cognitive components of CBT may not impart incremental validity over and above the behavioural components of CBT alone. In an initial study investigating this question, Jacobson and colleagues (Jacobson, Martell, & Dimidjian, 2001) randomized depressed adults to

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receive either the full package of CBT or to receive all of the behavioural components but none of the cognitive components of CBT. They found that both interventions were equally effective, both in the acute phase of treatment and in preventing relapse at 2-year follow-up (Jacobson et al., 2001). On the basis of these findings, Jacobson and colleagues developed the adult behavioural activation (BA) protocol, which has since been evaluated in additional randomized controlled trials (RCTs). In a recent RCT, Dimidjian and colleagues compared BA with cognitive therapy and antidepressant medication and found that BA was at least as effective as cognitive therapy (Dimidjian et al., 2006). Furthermore, they found that BA was comparable with antidepressant medication among severely depressed adults. Additionally, a recent meta-analysis of 34 studies of BA treatment for adults indicated a large and significant effect size for BA when compared with control conditions, and no differences in effectiveness were found between BA and cognitive therapy (Mazzucchelli, Kane, & Rees, 2009). Thus, BA represents a viable alternative to CBT for depressed adults and merits additional investigation with depressed adolescents.

Implementing BA as an alternative to cognitive therapy for adolescents is potentially advantageous based on variable developmental levels in children and adolescents. Some literature supports the idea that youth are better able to participate in interventions that are behaviorally rather than cognitively focused, regardless of developmental level (Wilkes, 1994). Thus, the development of behavioural interventions for depression stands to benefit a wide range of neurodevelopmental levels in depressed youth. Adapting the BA protocol for younger populations may fill a needed gap in the treatment literature for this vulnerable population.

Overview of Behavioural Activation

The primary goal of BA is to assist the client in implementing behavioural changes that increase the amount of positive reinforcement they encounter in their lives (Dimidjian et al., 2007). Behavioural activation was built on the theory that depressed individuals engage in behavioural patterns that provide few opportunities for environmental rewards. According to this theory, depressed individuals tend to withdraw from their environments in response to feelings of sadness and therefore adopt avoidant behavioural patterns that minimize the likelihood of both short- and long-term positive reinforcement. Therefore, the aim of BA is to disrupt the behavioural patterns that maintain depressive symptoms by creating a system of activities that increase positive activities and, therefore, opportunities for reward. The assignments in BA are tailored to the unique needs of the individual, and activities are chosen in collaboration with the therapist. The patient's progress is examined and discussed in each session, and modifications to the treatment plan are made as needed. In addition, the nature, course and duration of symptoms are closely monitored throughout therapy.

A recent pilot study of BA with depressed adolescents found that participants' depression scores decreased significantly over time; furthermore, participants' scores on a measure of psychological well-being improved significantly over 18 weeks of treatment (Ritschel et al., 2011). Other studies of BA with depressed adolescents are underway, but results have not yet been published; however, exploratory analyses comparing BA with treatment as usual have shown that attrition rates in the BA condition are low and that significant between-group differences have emerged that favour BA (McCauley, personal communication to WEC, 2010).

In our laboratory, we have adapted BA for adolescents and have collated our work into a manual that outlines six phases of treatment that were designed to be flexibly implemented (i.e., the treatment does not follow a strict session-by-session format; Ritschel et al., 2011). The first phase involves treatment orientation during which the BA model is reviewed with the patient. The second phase individualizes targets for behavioural activation and engagement. During this second phase, considerable time is spent examining and evaluating the contingencies that either serve to maintain or exacerbate depressive symptoms. Activity monitoring is a significant part of this phase of therapy, as it helps illustrate the relationship between certain behaviours (e.g., social interaction, participation in hobbies) and mood. The third phase focuses on improving problem-solving skills and approach-oriented coping skills as well as decreasing avoidant behaviour. The fourth phase involves instruction in long-term goal-setting, with a focus on breaking down goals into feasible steps in order to facilitate achievement and success. The fifth phase entails repeated application and modification of behavioural activation strategies. The final phase includes review of behavioural change and gains made during the course of treatment as well as a focus on relapse prevention. Parents are included at the beginning, middle and end of treatment and may be invited in to additional sessions at the discretion of the therapist and client. For example, parents have an active role in the orientation/commitment session so that they can understand and become invested in the treatment model. In addition, parents have the ability to ask question and provide input during non-designated parental sessions in order to make treatment a collaborative process. For example, parents may request assistance in resolving family conflict within the home environment. In this case, the therapist can engage the patient in using learned problem-solving skills (a BA intervention target of phase 3) to address problems with family members.

When adapting an existing treatment to account for important cultural and socio-economic factors, the flexibility of treatment is especially important, and this is a particular strength of BA adapted for adolescents. Behavioural activation is unique in its focus on environmental contingencies; consequently, this form of therapy has the capacity to take cultural and socio-economic factors into account. For example, previous research has shown that, in addition to more present-oriented, solution-focused approaches, psychotherapy with African American clients often includes some focus on spirituality or religion, as well as family relationships (Sue & Sue, 2003). The flexibility of BA allows for the incorporation of spiritual activities such as church involvement and prayer (outside of therapy) when selecting treatment targets and activities for behavioural activation. Additionally, through the inclusion of family sessions throughout treatment, BA allows for opportunities to build more frequent and positive family interactions, which are important given there is substantial research demonstrating that poor family cohesiveness is positively correlated with depression in African American adolescents (Herman, Ostrander, & Tucker, 2007). Behavioural activation in particular has been targeted in specific minority groups to address the environmental and social-economic stressors associated with various populations. Behavioural activation was helpful in treating depressed Latinos possibly because BA addressed immediate life circumstances and emphasized changing environmental variables that are identified as risk factors for this population (Kanter et al., 2010). Behavioural activation emphasizes 'perseverance, action, and empowerment in the face of seemingly overwhelming environmental hardships…' that may be more relevant to a Latino population (Kanter et al., 2010).

In sum, BA holds the potential to be an effective treatment for African American youth with MDD. In this case series, we conducted a preliminary evaluation of BA with three adolescent clients in order to evaluate the feasibility of implementing BA with this population and to determine if BA had an impact on the depression of these individuals. Participants were recruited from a large urban hospital that treats low-income, predominately African American minorities in the southeastern USA. Because of the preliminary nature of this study, we chose a case series design in which we established a basic set of inclusion and exclusion criteria to determine that the participants had relatively similar types of psychiatric difficulties; for example, all participants had to meet a minimum level of depression severity.

METHODS

Participants

A total of 10 African American subjects were screened for the study. Four subjects did not meet criteria for the study. Three subjects withdrew for reasons including starting medication, questionable psychosis or inability to make contact (no showed/contact information changed). Finally, two African American men (ages 14 and 15) and one African American woman (age 17) were included in this case series. Annual income for all participants' families was below $25 000, and all of the participants' primary caregivers were unemployed. Inclusion criteria were as follows: (a) principal diagnosis of MDD based on the Kiddie Schedule for Affective Disorders (K-SADS); (b) between 13 and 17 years of age; (c) Children's Depressive Rating Scale—Revised (CDRS-R) score ≥ 45 at baseline; and (d) Beck Depression Inventory—Second Edition (BDI-II) score ≥ 14 at baseline. Exclusion criteria were as follows: (a) current or past diagnosis of any psychotic spectrum disorder, pervasive developmental disorder, organic brain syndrome, substance abuse or dependence, or mental retardation; and (b) unwillingness to attend weekly sessions. Participants could be included if they were taking a psychotropic medication, provided that no medication changes had been made 2 months prior to initiation of BA treatment. If medication changes were made during treatment, then they would be noted on post-treatment analysis.

Measures

Diagnoses were established using the K-SADS (Kaufman et al., 1997), a clinician-administered, semi-structured interview, that is based on Diagnostic and Statistical Manual for Mental Disorders, Fourth Edition, diagnostic criteria and includes information obtained from both adolescent and parent reports. Depressive symptom severity was assessed with the CDRS-R (Pozanski & Mokros, 1996), BDI-II (Beck, Steer, & Brown, 1996) and Clinical Global Impressions—Severity scale (CGI-S) (National Institute of Mental Health, 1985). The CDRS-R is a 17-item semi-structured interview; ratings are based on adolescent report, parent report and clinician's combined interpretation. The CDRS-R has demonstrated high inter-rater reliability ($\alpha = 0.92$) (Pozanski & Mokros, 1996), good 2-week test–retest reliability (0.80) and very good internal consistency ($\alpha = 0.85$). Although we did not evaluate inter-rater reliability on these instruments, our assessor had been trained by watching detailed training videos as well as by direct observation. The BDI-II is a 21-item self-report measure that has demonstrated high internal consistency ($\alpha = 0.93$) and test–retest reliability (0.93 for a 1-week interval) (Beck et al., 1996). Studies of the BDI-II with adolescent outpatient samples have found similarly high internal consistency ($\alpha = 0.92$) with participants as young as age 13 (Steer et al., 1998). The CGI-S is a two-item, seven-point scale completed by the clinician that assesses depression severity ($0 =$ no illness, $6 =$ extremely ill) and symptomatic improvement across treatment. Although the CGI-S has shown only moderate psychometric validity, it is a widely used measure that has been shown to be of value in clinical drug trials provided that the rater knows his or her client (Guy, 1976). For this study, we also incorporated a post-treatment self-report measure of treatment satisfaction, which was given to both the child and the parent. Items assessed overall satisfaction with services received at the clinic, including likability (e.g., 'I liked going to the clinic') and acceptability of the treatment, as well as perceived progress (e.g., 'Going to the clinic helped me with my problems'). The child version used a four-point scale and the adult version incorporated a five-point scale, with higher scores indicating a greater satisfaction with services (Hawley & Weisz, 2005).

Procedure

Participants were recruited as part of the standard clinic intake process and by using flyers posted in local community mental health clinics. Potential participants completed a scripted telephone screen to determine preliminary eligibility. If the telephone screen criteria were met, the adolescent and his or her primary caregiver were invited to the clinic for an in-person baseline assessment, which included the K-SADS, CGI-S, CDRS-R and BDI-II. Participants who met inclusion criteria completed the BDI-II at each session. At approximately session 9 (approximately 9 weeks into treatment), adolescents and their primary caregivers also completed the CDRS-R at the midpoint of treatment. At the end of treatment (approximately 18–20 weeks after treatment began), participants and their primary caregivers completed a post-treatment assessment consisting of the K-SADS, CGI-S, CDRS-R, BDI-II and the Treatment Satisfaction Questionnaire—Teen and Parent versions. Of note, because of the lack of resources and staff at our site, our primary clinician administered both the pre- and post-assessments. Although this clearly added potential bias to the clinically rated outcomes, we felt that the clinical relevance of gathering pre- and post-treatment trends outweighed the barriers of this type of methodology and reasons to not attempt BA at all. Therefore, readers should understand these issues when interpreting the results.

Treatment

As mentioned earlier, BA can be construed as consisting of six general phases, as the treatment does not follow a strict session-by-session format. The first phase involves orientation to treatment, in which the BA model is reviewed with the patient (sessions 1–3). Here, the therapist describes the BA model, illustrates the upward spiral and downward spiral of feelings and behaviours to the patient and orients the patient's caregiver to the model so that they can become co-collaborators in the treatment process. During the first three sessions, various handouts are completed so that the patient can incorporate his or her own examples of life circumstances that are consistent with the BA model. In addition, since the therapy is flexible, techniques that are scheduled to be taught in future sessions can be brought up early if patients are dealing with crises in the earlier sessions. For example, some patients may need help with problem solving (taught in a later phase of treatment) if a family conflict arises within the first three sessions.

The second phase involves individualizing targets for behavioural activation and engagement (sessions 4–6). During this phase, the patient and the therapist evaluate the contingencies that either maintain or exacerbate depressive symptoms. During the first few sessions of this phase, activity monitoring charts are introduced in session, and the patient is encouraged to start recording activities and the moods associated with each activity in order to understand the connection between the two. Subsequent sessions in this phase focus on brainstorming activities that provide pleasure and mastering and building these activities into the patients' daily schedule. Patients are then asked to begin a worksheet for planning their day by identifying specific tasks in which they can partake.

The third phase involves focus on instruction in problem-solving skills as well as approach-oriented coping skills (sessions 7–8). In this phase, the patient discusses their various avoidance strategies and acknowledges that these strategies have not helped with the negative emotions. The problem-solving strategies taught in this phase are based upon the 'ACTION' acronym that teaches patients to Assess their current behaviour, Choose an alternative behaviour if the current behaviour does not work for them, Try the behaviour out and continue to Integrate and Observe the results of the New behaviour. In addition, the therapist teaches skills, such as mindfulness, aimed at reducing rumination.

The fourth phase involves instruction in long-term goal-setting skills, with a focus on breaking down goals into feasible steps in order to facilitate achievement and success (sessions 9–10). The patient is taught to identify short-term goals and long-term goals. They complete handouts asking them to break down each goal into smaller sub-goals until they finally achieve their larger goal. In this phase, the patient may use mental rehearsal to help them practice their goals.

The fifth phase entails repeated application and modification of behavioural activation strategies (11–16). In this phase, contingencies are developed in order to complete BA tasks. In addition, the 'TRAP and TRAC' acronyms are discussed where the patient identifies an 'environmental trigger', an 'emotional reaction' and an 'avoidance pattern'; the patient then uses the same trigger and response but is encouraged to develop an 'alternative coping' strategy. These concepts are practiced repeatedly along with other techniques used in previous phases.

The final phase includes review of behavioural changes and gains made during the course of treatment as well as a focus on relapse prevention (sessions 17–18). This phase involves reviewing skills during the course of BA therapy (and formulating a BA toolkit including coping cards, reminder notes etc), identifying themes that emerged during therapy and identifying when a patient my need more help for possible recurrent depression. The patient will also be encouraged to identify future stressors and how to deal with them as well as give feedback to the therapist about BA therapy. During all sessions, the therapist and patient set an agenda, do homework review and assign new homework.

Training and Therapists

Participants were invited to attend 14–17 BA sessions over a 6-month period. The treatment was based on the protocol reported by Ritschel and colleagues (Dobson et al., 2008). The first, third and fourth authors completed BA training in 2007 during a intensive BA workshop conducted by Dr Sona Dimidjian, one of the originators of BA. Intensive and ongoing individual and group supervision occurred in a weekly BA study group at the Emory Child and Adolescent Mood Program.

OVERALL RESULTS

Table 1 shows the rates of no-shows and cancellations for all three participants. Although these rates were fairly high, they are consistent with research documenting the high rates of no-shows and cancellations for mental health service use for youth (McKay, McCadam, & Gonzales, 1996). Pre- and post-treatment scores on all outcome measures (CDRS-R, BDI-II, CGI-S and K-SADS) are shown in Table 2. At post-treatment, two of the three participants had CDRS-R scores in the 'normal' range, and all three participants' BDI scores were within normal limits at post-treatment. Clinician-rated CGI and K-SADS scores indicated that two of the three participants were no longer depressed at post-treatment, and the third participant was rated as still clinically depressed, with moderate levels of depression severity. In addition, the results of the changes in BDI-II scores over treatment phases are shown in Table 3.

Table 1. Rate of no-shows and cancellations throughout treatment

Subject	No-show	Cancellation
1	2	6
2	5	1
3	1	8

Satisfaction ratings showed that participants and their caregivers were satisfied to very satisfied with treatment. The overall mean youth satisfaction rating was 10.7 (of 12; range = 10–11), and the overall mean caregiver satisfaction rating was 18.7 (of 20; range = 18–19).

INDIVIDUAL RESULTS

Teen 1

(Please note that all names have been changed to protect confidentiality.)

Derrick, a 14-year-old boy who met criteria for moderately severe MDD (CDRS = 45 and BDI = 21) and Attention Deficit Hyperactivity Disorder (ADHD) at baseline, completed 17 sessions of BA. Although Derrick regularly attended sessions on a weekly basis, several sessions were missed during the third phase of treatment due to family financial difficulties, as there were concerns that the family would be evicted from their home. The onset of Derrick's depressive episode began following his family's relocation to a new apartment complex due to eviction from their previous home, resulting in a change of schools during the middle of the academic year. His mother reported that he began withdrawing from social activities and 'having an attitude all the time'. During the baseline assessment, Derrick endorsed symptoms of depressed mood, irritability, anhedonia, psychomotor agitation, poor appetite, lack of energy and guilt feelings. He also endorsed engaging in avoidant behaviours in order to cope with stress, such as staying in his room all day and not hanging out with friends after school. His treatment goals were 'to make more friends, play sports again, and make better decisions'. His mother, who participated in the orientation session, agreed with Derrick's treatment goals and was very supportive of his involvement in treatment. Consistent with the BA approach, the early stages of his treatment focused on evaluating the contingencies that were maintaining his depressive symptoms. By completing activity monitoring worksheets, he was able to identify the relationship between avoidant behaviours (e.g., staying in his room all day) and mood (e.g., increased feelings of depression and/or irritability). Following his increased awareness of the connection between behavioural choices and subsequent moods, treatment focused on identifying both pleasurable and mastery building activities that he could systematically incorporate into his daily schedule. Using problem-solving and goal-setting activities, he and his therapist identified a strategy to help him join the school football team, and he was able to

Table 2. Participant demographics, diagnoses and nature of symptoms

Subject	Gender	Age	Diagnosis	CDRS-R total		BDI-II total		CGI-S		K-SADS	
				Pre	Post	Pre	Post	Pre	Post	Pre	Post
1	Male	14	MDD; ADHD	45	22	21	4	4	1	Yes	No
2	Male	15	MDD; DD	72	55	27	6	6	4	Yes	Yes
3	Female	17	MDD; SP	61	22	17	2	4	1	Yes	No
			Mean (SD)	59.3 (13.6)	33.0 (19.1)	21.7 (4.1)	4.0 (2.0)	4.7 (1.2)	2.0 (1.7)		

SD = standard deviation. CDRS-R total = Children's Depression Rating Scale—Revised total score. BDI-II total = Beck Depression Inventory—Second Edition total score. CGI-S = Clinical Global Impression—Severity rating. K-SADS = Kiddie Schedule for Affective Disorders. MDD = major depressive disorder. ADHD = attention deficit hyperactivity disorder. DD = dysthymic disorder. SP = social phobia.

follow through with this plan successfully. During his participation in football, he worked with his therapist to develop the skills he needed to establish friendships with several teammates, who invited him to play basketball and go skateboarding with them after football season ended. He noticed that his mood improved when he spent time engaging in these activities with his friends, and consequently, he spent less time isolating himself. As treatment progressed, he and his therapist worked on identifying concrete goals related to his academic, family and athletic functioning. They worked to break large goals down into specific and attainable sub-goals. For example, Derrick's goal of achieving honour roll was broken down into specific steps, such as attending tutoring sessions and creating study guides for tests. His mother requested a specific session to address a family-related problem in which she was concerned that Derrick was taking personal items from her without her permission. During the session, Derrick was able to use learned problem-solving skills to identify the problem (i.e., taking things without asking) and to develop a solution that would conflict with his mother (i.e., ask permission before taking things, do chores around the house to earn money in order to buy needed things). Through this process, Derrick was able to work systematically toward achieving his goals that resulted in enhanced mood and self-esteem.

Certain aspects of Derrick's family's socio-economic status played an important role in Derrick's treatment. For example, during a mid-treatment session aimed at discussing with his mother options for enhancing Derrick's involvement in enjoyable activities, his mother expressed that she was not comfortable with him skateboarding with friends outside due to safety risks, as his mother reported that crime and drug use was common in their low-income neighbourhood. During BA sessions, time was spent problem-solving alternative areas for Derrick to engage in outdoor activities, such as at his school and at a park nearby his house, where there was adequate adult supervision and relatively lower safety risks. Additionally, in order to engage his mother further in treatment, the therapist used case management skills to locate resources in the community, including the Boys and Girls Club, which would provide additional supervision for Derrick. His mother responded very well to this additional support and appeared willing to allow Derrick to take advantage of community resources.

By the end of treatment, Derrick had demonstrated significant progress toward achieving his goals for therapy. He reported improved mood and motivation to participate in social and athletic activities, enhanced problem-solving skills and increased self-confidence. Moreover, his teachers and classmates voted him as 'Student of the Month' based on his academic performance. At the end of treatment, his mother reported that he seemed happier at home, and she indicated that he fought less with his brothers. At post-treatment assessment, Derrick no longer met criteria for MDD based on the K-SADS. His post-treatment BDI-II score was 4 and his post-treatment CDRS-R score was 22.

Teen 2

Greg, a 15-year-old boy who met criteria for severe MDD and dysthymic disorder at baseline, attended 15 sessions of BA (CDRS = 72 and BDI = 27). Greg consistently attended sessions for all phases of treatment with the exception of the last phase of treatment. He no-showed for approximately three weeks, and he nor his parents returned telephone calls. At the end of treatment, he reported that his no-shows were related to fears of ending treatment. The onset of his depressive symptoms was traced back 9 years to the death of his grandmother, who had served as his primary caregiver. Following this event, Greg withdrew from social activities and 'became an

Table 3. Average Beck Depression Inventory—Second Edition scores across treatment phases

Subject	Phase 1	Phase 2	Phase 3	Phase 4	Phase 5	Phase 6
1	15	11	8	11	8.17	n/a
2	22	11	7	5	12	5.6
3	12	9	8	6	3.25	n/a

'n/a' marked when participants did not attend sessions 17 and 18.
BDI-II = Beck Depression Inventory—Second Edition.

outcast'. Throughout his life, he experienced numerous psychosocial stressors that maintained and exacerbated his depressive symptoms, including his sister's relocation to another state, witnessing domestic violence between his mother and father, enduring the sequelae of his parents' substance abuse and suffering from neglect. At the baseline assessment, Greg endorsed symptoms of depressed mood, irritability, anhedonia, psychomotor agitation, hypersomnia (>14 hours sleep per day), increased appetite, fatigue, poor self-esteem, poor concentration, hopelessness and passive suicidal ideation. His treatment goals were 'be less tired and make more friends'.

Greg had a difficult time engaging in treatment due in part to his substantial neurovegetative symptoms and attentional difficulties. In their early work together, his therapist devoted a considerable amount of time to building rapport and to orienting the client to the BA model of treatment. Over time, Greg was able to express an understanding of the BA model, and he was able to begin to work toward his treatment goals. In order to increase his social activity, he began by joining the chess club at school, and he reported that this helped improve his mood. He noted that his mood only lifted while he was actively engaged in club events, and his depressive symptoms returned for the remainder of the day. Several factors related to Greg's socio-economic status resulted in significant obstacles to therapeutic success. For example, Greg's family did not have enough income to pay for extracurricular activities; they could not afford to enrol him in a drawing class or buy him a chess board. Additionally, due to lack of access to regular transportation, Greg was not able to visit his cousins, with whom he enjoyed spending time and who lived less than 10 miles away, because their residence was not near a bus line. Furthermore, although Greg was interested in volunteering at an animal shelter, he was not able to participate in this activity because his parents (who had substantial substance abuse issues) had difficulty following through with signing consent forms and attending required parent orientation seminars. Finally, child protective services (CPS) was involved with Greg's family throughout treatment due to suspicions of neglect, and on three occasions, the therapist initiated calls to CPS due to concerns related to lack of food and inadequate parental involvement in therapeutic services. Despite numerous attempts to engage his parents in treatment, his parents did not come to appointments and frequently did not return telephone calls. Eventually, CPS became helpful in providing some transportation services so that Greg could come to appointments, and they checked in with Greg and his family to ensure that Greg's basic needs were met. Thus, there were many barriers related to Greg's family and socio-economic circumstances that affected therapeutic progress within the BA model. Although his mother and father participated in the baseline assessment and his mother participated in the post-treatment assessment, they did not attend therapeutic sessions. At the end of treatment, he continued to meet criteria for MDD and dysthymic disorder according to the CDRS-R (CDRS = 55); however, Greg's BDI-II scores suggested some symptomatic improvement between pre- and post-treatment (BDI-II = 6 at post-treatment).

Teen 3

Adrian, a 17-year-old girl who met criteria for severe MDD and social phobia at baseline, attended 14 sessions of BA (CDRS = 61 and BDI = 17). Her session attendance was more sporadic, in part due to family and transportation problems. Specifically, the family car broke down for approximately 3 weeks. Additionally, due to financial problems, her sister had to cancel sessions because she did not have enough money to pay for gas. The onset of Adrian's depressive symptoms had begun approximately 2 years earlier; this onset followed an incident at school in which she was ridiculed for her appearance in front of her classmates. She was hospitalized for suicidal ideation on three separate occasions following this incident, as she reported that her self-esteem progressively worsened as peers continuously bullied her. Following hospitalization discharges, she began skipping school and frequently arguing with her mother; these events culminated in Adrian moving to the Atlanta area to live with her older sister. At the baseline assessment, Adrian endorsed symptoms of depressed mood, irritability, anhedonia, hyposomnia, excessive appetite, poor concentration and feelings of guilt and worthlessness. Her treatment goals were to 'stay in school, be more social, and be less angry'. Her sister, who was involved in the orientation session, agreed with these goals and indicated a specific desire for Adrian to be more involved in school-related activities rather than 'staying in her room all day'. Adrian was very engaged in treatment and came to each session with specific issues to address. A large part of treatment was devoted to instruction and practice in problem-solving skills to cope with psychosocial stressors, such as conflict with her sister and teasing by her peers. Her sister attended several problem-solving sessions aimed at resolving conflict at home. During one of the sessions, her sister described an interest in receiving her own individual therapy.

Throughout treatment, Adrian benefited from participating in exercises that traced the connections between feelings and behaviours, and she learned how to change a 'downward spiral' of negative events and feelings into a 'positive spiral' by actively choosing an approach-oriented coping strategy. For example, instead of retreating to her room after a disappointing day at school, she chose to discuss her feelings with her sister, who in turn provided emotional support that led to improved mood. Due to her sister's active involvement in treatment, her sister sought the support from the therapist via telephone contact in order to ensure that she was responding to Adrian appropriately.

At times, Adrian's sister was unable to attend sessions due to financial constraints on transportation (e.g., car breaking down, not enough money to pay for gas) as well as her own struggles with depression. In order to overcome these barriers, telephone sessions were utilized on occasion in order to continue to engage Adrian in treatment. Additionally, the therapist provided Adrian's sister with low-cost referrals so that she could also seek psychological treatment.

At post-treatment, Adrian reported improved problem-solving skills and increased interest in engaging in social activities. She had made several new friends at school, and she was enjoying spending more time with them. Her sister reported progress in Adrian's approach-oriented coping skills and indicated a decrease in her depressive and anxiety symptoms. At the end of treatment, she no longer met criteria for MDD or social phobia. Her post-treatment BDI-II score was 2, and her post-treatment CDRS-R score was 22.

DISCUSSION

Results from this case series provide preliminary support for BA as a treatment for low-income, African American adolescents with MDD. Two of the three participants were classified as being in remission at post-treatment, as they no longer met criteria for MDD based on clinician report and assessor interview. Furthermore, all participants and caregivers reported satisfaction with the treatment.

We note several points regarding why BA may have been effective in this low socio-economic status, African American group of adolescents with MDD. First, patients and their families were able to use community activities and resources to help achieve behavioural goals that have been described to be culturally important in low socio-economic status patient groups (Anderson et al., 2003). Participants were able to understand the core components of BA and appropriately implement these principles in their daily lives. For example, Greg joined the chess club at school and was considering enlisting in job corps when he became of age. Second, BA's focus on increasing pleasurable activities appeared to be a motivator for participants' engagement in treatment, which has historically been a problem exhibited in patients in this population. Participants engaged in BA treatment as evidenced by their enthusiastically brainstorming ideas for behavioural activation during treatment sessions. In two of the three cases, participants' family members were also engaged in BA treatment and, in particular, benefitted from involvement in problem-solving exercises aimed at reducing family conflict. Additionally, objective data from a treatment satisfaction measure indicated that participants liked coming to therapy. Finally, participants appeared to gain practical skills that were directly transferable to real-world goals. For example, Adrian requested that she practice interview skills with her therapist so that she would feel more prepared to achieve her goal of obtaining a summer job, and Greg spent several sessions practicing his chess skills for about 15 minutes during therapy so that he felt more confident for his weekly chess club meetings.

Of course, these results may be due to factors other than the BA intervention. For example, we allowed subjects to continue psychotropic medications provided that the dose had not been changed 2 months prior to the study and the medications were not changed during the course of therapy. In this case series, one patient (Derrick) was prescribed a low-dose antidepressant medication (trazadone 100 mg) for sleep difficulties as well as a stimulant to treat ADHD (methylphenidate extended release 54 mg daily plus methylphenidate 10 mg in the afternoon). These dosages had been stable for several months prior to the study. Thus, although the medications alone were not effective for remediating his depression, it may have been the interaction of BA and medication that produced the substantial reductions in depressive affect that were observed at post-treatment.

As expected of any therapeutic intervention, BA is not effective for all depressed teens. Specifically, Greg did not benefit from the treatment to the same extent as the other two participants. There are multiple ways in which to interpret these findings. For example, BA may be ineffective for (1) treating comorbid depression and dysthymia (i.e., 'double depression'); (2) an adolescent patient whose family life is sufficiently chaotic to warrant the need for CPS; (3) treating a patient who suffers from severe neurovegetative symptoms; or (4) parents who are not adequately involved in the therapeutic process.

The present case series had several methodological limitations. First, given the 'open label' nature of this trial, improvement may be attributable to nonspecific treatment factors (e.g., attention), the passage of time or regression to the mean. Second, although much attention was given to treatment engagement and compliance with sessions, perhaps more attention should be given in this area initially to help attendance and completion of tasks for every session. For example, more telephone sessions could have occurred in addition to requiring more family sessions. Additionally, the second author served both as the assessor and therapist for each participant, and therefore, the results may be unintentionally biased. At a minimum, the trends in outcome suggest that BA in this group of adolescents is worth further examination. What appears evident is that self-report measures were generally comparable with clinician-rated assessments. Furthermore, this case series did not incorporate a follow-up assessment beyond the post-test to evaluate whether symptom improvements were sustained. Future study of this approach should incorporate structured follow-up assessments. Finally, future studies should evaluate potential moderators of therapeutic outcomes to identify both clinical (e.g., comorbidity, symptom severity) and demographic (e.g., socioeconomic status, family composition) factors that may affect therapeutic success. Within these limitations, our data suggest that BA is feasible and has the potential of great clinical utility in reducing depressive symptoms in low-income, African American youth.

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REFERENCES

Anderson, L., Scrimshaw, S., Fullilove, M., Fielding, J., & the Task Force on Community Preventive Services. (2003). The community guide's model for linking the social environment to health. *American Journal of Preventive Medicine*, *24*, 12–20.

Beck, A.T., Steer, R.A., & Brown, G.K. (1996). *Beck Depression Inventory II: Manual*. New York, NY: Psychological Corporation.

Birmaher, B., Brent, D., & the AACAP work group on quality issues. (2007). Practice parameters for the assessment and treatment of children and adolescents with depressive disorders. *Journal of the American Academy of Child and Adolescent Psychiatry*, *46*, 1503–1526.

Brent, D.A. (2001). Assessment and treatment of the youthful suicidal patient. *Annals of the New York Academy of Sciences*, *932*, 106–128.

Dimidjian, S., Hollon, S.D., Dobson, K.S., Schmaling, K.D., Kohlenberg, R.J., Addis, M.E., & Jacobson, N.S. (2006). Randomized trial of behavioral activation, cognitive therapy, and antidepressant medication in the acute treatment with adults with major depression. *Journal of Consulting and Clinical Psychology*, *74*, 658–670.

Dimidjian, S., Martell, C.R., Addis, M.E., & Herman-Dunn, R. (2007). Behavioral activation for depression. In D.H. Barlow (Ed.), *Clinical handbook of psychological disorders* (pp. 328–364). New York, NY: The Guilford Press.

Dobson, K.S., Hollon, S.D., Dimidjian, S., Schmaling, K.B., Kohlenberg, R.J., Gallop, R., & Jacobson, N.S. (2008). Randomized trial of behavioral activation, cognitive therapy, and antidepressant medication in the prevention of relapse and recurrence in major depression. *Journal of Consulting Clinic Psychology*, *76*,468–477.

Guy, W. (1976). *ECDEU assessment manual for psychopharmacology* (pp. 218–222). Rockville, MD: National Institute for Mental Health.

Hankin, B.L., Abramson, L.Y., Moffitt, T.E., Silva, P.A., McGee, R., & Angell, K.E. (1998). Development of depression from preadolescence to young adulthood: Emerging gender differences in a 10-year longitudinal study. *Journal of Abnormal Psychology*, *107*, 128–140.

Hawley, K.M., & Weisz, J.R. (2005). Youth versus parent working alliance in usual clinical care: Distinctive associations with retention, satisfaction, and treatment outcome. *Journal of Clinical Child and Adolescent Psychology*, *34*, 117–128.

Herman, K.C., Ostrander, R.O., & Tucker, C.M. (2007). Do family environments and negative cognitions of adolescents with depressive symptoms vary by ethnic group? *Journal of Family Psychology*, *21*, 325–330.

Jacobson, N.S., Martell, C.R., & Dimidjian, S. (2001). Behavioral activation for depression: Returning to contextual roots. *Clinical Psychology: Science and Practice*, *8*, 255–270.

Kann, L., Kinchen, S.A., Williams, B.I., Ross, J.G., Lowry, R., Grunbaum, J.A., & Kolbe, L.J. (2000). Youth risk behavior surveillance—United States, 1999. State and local YRBSS coordinators. *Journal of School Health*, *70*, 271–285.

Kanter, J.W., Azara, L.S.-R., Rusch, L.C., Busch, A.M., & West, P. (2010). Initial outcomes of a culturally adapted behavioral activation for Latinas diagnosed with depression at a community clinic. *Behavior Modification*, *34*(2), 120–144.

Kaufman, J., Birmaher, B., Brent, D., Rao, U., Flynn, C., Moreci, P., & Ryan, N. (1997). Schedule of affective disorders and schizophrenia for school-age children—Present and lifetime version (K-SADS-PL): Initial reliability and validity data. *Journal of the American Academy of Child and Adolescent Psychiatry*, *36*, 980–988.

Kessler, R.C., & Walters, E.E. (1998). Epidemiology of DSM-III-R major depression and minor depression among adolescents and young adults in the National Comorbidity Survey. *Depression and Anxiety*, *7*, 3–14.

Klein, J.B., Jacobs, R.H., & Reinecke, M.A. (2007). Cognitive-behavioral therapy for adolescent depression: A meta-analytic investigation of changes in effect-size estimates. *Journal of the American Academy of Child and Adolescent Psychiatry*, *46*, 1403–1413.

Lewinsohn, P.M., & Clark, G.N. (1999). Psychosocial treatments for adolescent depression. *Clinical Psychology Review*, *19*, 329–342.

Lewinsohn, P.M., Rohde, P., & Seeley, J.R. (1998). Major depressive disorder in older adolescents: prevalence, risk factors, and clinical implications. *Clinical Psychology Review*, *18*, 765–794.

Mazzucchelli, T., Kane, R., & Rees, C. (2009). Behavioral activation treatments for depression in adults: A meta-analysis and review. *Clinical Psychology: Science and Practice*, *16*, 383–411.

McKay, M., McCadam, K., & Gonzales, J. (1996). Addressing the barriers to mental health services for inner city children and caretakers. *Community Mental Health*, *32*, 353–361.

Michael, K.D., & Crowley, S.L. (2002). How effective are treatments for child and adolescent depression? A meta-analytic review. *Clinical Psychology Review*, *22*, 247–269.

National Institute of Mental Health. (1985). Rating scales and assessment instruments for use in pediatric psychopharmacology research. *Psychopharmacology Bulletin*, *21*, 839–843.

Pozanski, E.O., & Mokros, H.B. (1996). *Children's depression rating scale—Revised manual*. Los Angeles, CA: Western Psychological Services.

Ritschel, L.A., Ramirez, C., Jones, M., & Craighead, W.E. (2011). Behavioral activation for depressed teens: Results of a pilot study. *Cognitive and Behavioral Practice*, *18*, 281–299.

Ryan, N.D. (2005). Treatment of depression in children and adolescents. *The Lancet*, *366* (9489), 933–941.

Steer, R., Kumar, G., Ranieri, W., & Beck, A. (1998). Use of the Beck Depression Inventory-II with adolescent psychiatric outpatients. *Journal of Psychopathology and Behavioral Assessment*, *20*(2), 127–137.

Sue, D.W., & Sue, D. (2003). *Counseling the culturally diverse: Theory and practice*. New York, NY: John Wiley & Sons.

U.S. Food and Drug Administration (USFDA). (2007). Anti-depressant use in children, adolescents, and adults. Retrieved August 2, 2011, from USFDA website http://www.fda.gov/drugs/drugsafety/informationbydrugclass/ucm096273.htm

Wilkes, T.C.R. (1994). Developmental Considerations. In T.C.R. Wiles, G. Belsher, A.J. Rush, & E. Frank and associates (Eds), *Cognitive therapy with depressed adolescents* (pp. 69–79). New York, NY: Guilford Press.

World Health Organization (WHO). Suicide prevention. Retrieved October 19, 2010, from http://www.who.int/menal_health/prevention/suicide/suicideprevent/en/

